# C1.3 Photosynthesis: data-based questions

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 13 |

**Read the data carefully before you answer.** The number of marks shows how many separate points to make. Show your working in calculations.

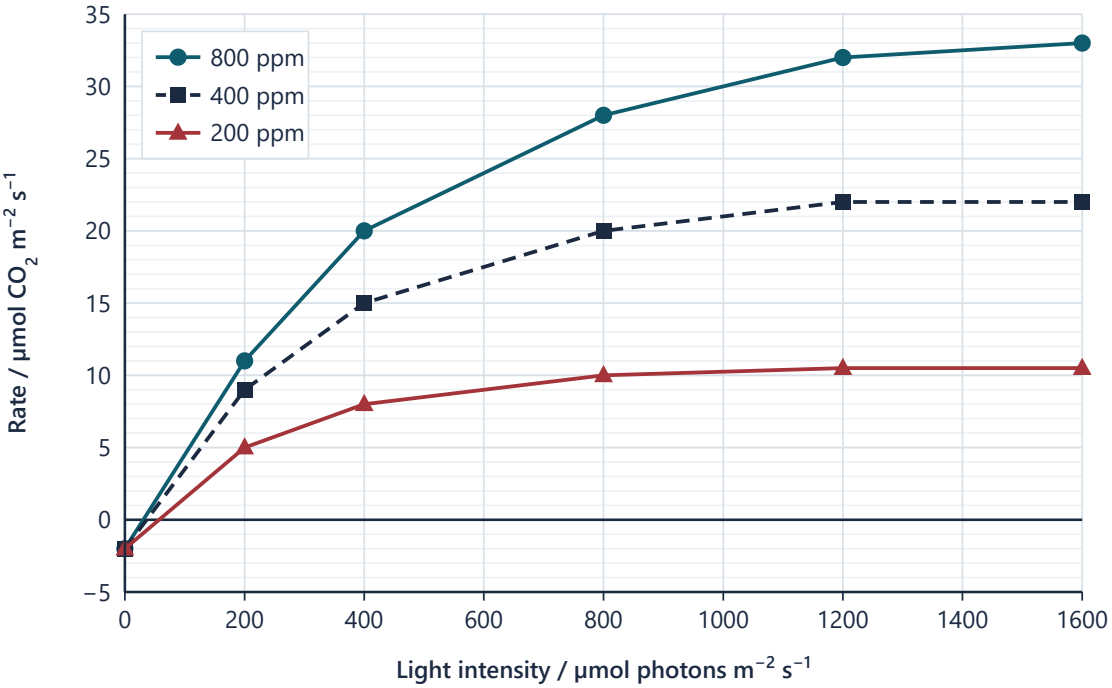
## 1 Light, carbon dioxide and wheat

8 MARKS



Image: Jon Sullivan, Public domain, Wikimedia Commons.

Wheat seedlings were grown at three carbon dioxide concentrations, in parts per million (ppm). The rate of photosynthesis was measured as net carbon dioxide uptake at different light intensities. Temperature was kept at 25 °C.




**a** STATE the rate of photosynthesis at 800 µmol photons $m^{-2}$ $s^{-1}$ and 400 ppm carbon dioxide. [1]

**b** OUTLINE the effect of carbon dioxide concentration on the rate of photosynthesis. [3]

c IDENTIFY the factor limiting the rate at 1600 μmol photons $m^{-2}$ $s^{-1}$ for the 400 ppm plants. Give a reason. [1]

d EXPLAIN why the rate is negative in darkness. [1]

e CALCULATE the percentage increase in the highest rate when carbon dioxide is raised from 400 to 800 ppm. [1]

f SUGGEST one reason why growers may add carbon dioxide to greenhouses only when light is bright. [1]

## 2 Sun and shade leaves

5 MARKS

Leaves of English ivy (*Hedera helix*) that grew in full sun were compared with leaves that grew in the shade of other leaves on the same plant. Values are means of 20 leaves, ± 1 SD.

| | SUN LEAVES | SHADE LEAVES |
|---|---|---|
| LEAF AREA / $CM^2$ | 18 ± 3 | 31 ± 5 |
| TOTAL CHLOROPHYLL / MG $G^{-1}$ | 1.6 ± 0.2 | 2.4 ± 0.3 |
| CHLOROPHYLL A : CHLOROPHYLL B | 3.4 | 2.6 |

a COMPARE the area of sun and shade leaves. [1]

b CALCULATE how much more chlorophyll per gram the shade leaves contain, as a percentage. [1]

c EXPLAIN how a larger area and more chlorophyll help shade leaves. [2]

d Chlorophyll b absorbs some wavelengths that chlorophyll a absorbs poorly. SUGGEST why shade leaves have a lower ratio of chlorophyll a to b. [1]